

Computational Science M.S. Thesis Proposal

Production of massless spin-0 particles in analog (3+1)-dimensional de Sitter universes from a ground state BEC using high-performance computing

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The story so far - Analog Universes

- ① Big Bang and rapid expansion $\implies$ could observe particles
- ② Big Bang was 14 billion years ago $\implies$ we can't observe particles
- ③ Condensed matter dynamics match particle dynamics (see Methods)
- ④ Can make condensed matter in the lab $\implies$ can effectively observe Big Bang particles
- ⑤ Lab based condensed matter system = “analog universe”

The story so far - Previous Research

Previous research has been done investigating a variety of different systems

- ① “Hyper-phononic” quasiparticles and Lorentz Violation
- ② Analog massive particles
- ③ Analog particles with non-zero quantum spin
- ④ Analog universes with different expansion behavior
- ⑤ Analog universes with non-zero background velocity
- ⑥ Experimental realizations (some extremely relevant and recent)

The story so far - 2D & Low-Performance Computing

- 1 Analog $(2+1)$ -dimensional universe with massless, scalar particles in a de Sitter expansion has been investigated
- 2 Previous investigation provides theory and guide for conceptual analysis
- 3 3d analog universe needs high-performance computing because of increased size of wave vector domain (see Methods)
- 4 Without 3d, 2d error is unknown

My story

My focus is on

- ① $(3+1)$ D analog universe with high-performance computing
- ② linear dispersion & phononic quasiparticle
- ③ analog massless particles
- ④ analog spin-0 particles
- ⑤ analog de Sitter expansion
- ⑥ no background velocity

Goals

- ① Provide insight into the dynamics of particles created due to the expansion of the universe in the moments just after the Big Bang.
- ② Quantitatively describe the error in using a reduced-dimensional physical model as compared to a full-dimensional model.
- ③ Quantitatively describe the efficiency of using a full-dimensional model with high-performance computing.
- ④ Provide an open-access, “ready to use”, packaged, high-performance computational model.

Methods Overview

① Physical Theory

- ① Setting up the Analog
- ② Speed of Sound & Universe Expansion
- ③ The Field Equation in Context
- ④ Particle Production
- ⑤ Computational Parameters

② Computation

- ① Computational Algorithms
- ② Programming Language

Methods - Physical Theory - Setting up the analog

- 1 Tangible: Ground state Bose-Einstein condensate with no thermal or quantum fluctuations
- 2 Mathematical: Gross-Pitaevskii equation (GPE) under a Bogoliubov mean-field approximation (aka the *nonlinear Schrödinger equation*),

$$i\hbar\partial_t\psi(\mathbf{x}, t) = \left[-\frac{\hbar^2}{2m}\nabla^2 + V_{\text{ext}}(x) + U|\psi(\mathbf{x}, t)|^2 \right] \psi(\mathbf{x}, t). \quad (1)$$

Methods - Physical Theory - Setting up the analog

- 1 Substitute the linearized Madelung density-phase representation (below) into eq. (1)

$$\psi(\mathbf{x}, t) = \sqrt{n_0 + \hat{n}(\mathbf{x}, t)} e^{i(\theta_0 + \hat{\theta}(\mathbf{x}, t))} \quad (2)$$

- 2 Yields the phase field equation in d spatial dimensions ($g := \det g_{\mu\nu}$)

The Field Equation

$$\frac{1}{\sqrt{-g}} \partial_\mu \left[\sqrt{-g} g^{\mu\nu} \partial_\nu \hat{\theta} \right] = 0 \quad g_{\mu\nu} = \left(\frac{n_0}{c(t)} \right)^{2/(d-1)} \begin{bmatrix} -c(t)^2 & \vdots & 0 \\ \dots & & \dots \\ 0 & \vdots & \delta_{ij} \end{bmatrix} \quad (3)$$

Methods - Physical Theory - Setting up the analog

The Analog

The field equation describes the dynamics of

- the phase of collective oscillations in the BEC
- particles produced due to the expansion of the universe

The Field Equation

$$\frac{1}{\sqrt{-g}} \partial_\mu [\sqrt{-g} g^{\mu\nu} \partial_\nu \hat{\theta}] = 0 \quad g_{\mu\nu} = \left(\frac{n_0}{c(t)} \right)^{2/(d-1)} \begin{bmatrix} -c(t)^2 & \vdots & 0 \\ \dots & & \dots \\ 0 & \vdots & \delta_{ij} \end{bmatrix}$$

Methods - Physical Theory - Speed of Sound & Universe Expansion

Speed Analog

Speed of sound in BEC = $c(t)$ = Speed of light in analog universe

- 1 Define $c(t)$ in the context of BEC

$$c(t)^2 = \frac{U(t)n_0}{m} = \frac{4\pi\hbar^2}{m^2}n_0a(t) \quad (4)$$

- 2 Scaling function $b(t)$

$$\overbrace{U(t) = U_0 b(t)}^{\text{BEC Interaction Strength}} \implies \overbrace{c(t) = c_0 \sqrt{b(t)}}^{\text{Analog Universe Light Speed}} \quad (5)$$

Methods - Physical Theory - Speed of Sound & Universe Expansion

- A decreasing speed of light is indistinguishable from an expanding universe (i.e. $c'(t) \propto 1/\text{expansion}$)

Expansion Analog

A decreasing BEC interaction strength yields an expanding analog universe

- Can tune interaction strength in the lab to induce a specific expansion
- $b(t)$ defines the expansion

Methods - Physical Theory - The Field Equation in Context

- 1 Explicitly expansion dependent phase field equation in (2+1)D (left) and conformal time (right)

$$\partial_\eta^2 \hat{\theta} - \frac{1}{2} \frac{b'(\eta)}{b(\eta)} \partial_\eta \hat{\theta} - c_0^2 \nabla^2 \hat{\theta} = 0 \quad d\eta = \sqrt{b(t)} dt \quad (6)$$

- 2 Field Fourier Expansions in wave vector $\mathbf{k}$

$$\hat{\theta}(\mathbf{x}, t) = \frac{1}{\sqrt{V}} \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{x}} \chi_{\mathbf{k}}(t) \hat{b}_{\mathbf{k}}(t) + e^{-i\mathbf{k} \cdot \mathbf{x}} \chi_{\mathbf{k}}^*(t) \hat{b}_{\mathbf{k}}^\dagger(t) \quad (7a)$$

$$\hat{n}(\mathbf{x}, t) = \frac{1}{\sqrt{V}} \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{x}} n_{\mathbf{k}}(t) \hat{b}_{\mathbf{k}}(t) + e^{-i\mathbf{k} \cdot \mathbf{x}} n_{\mathbf{k}}^*(t) \hat{b}_{\mathbf{k}}^\dagger(t) \quad (7b)$$

Methods - Physical Theory - The Field Equation in Context

- ③ Choose de Sitter expansion for early universe (t_s is the “rate of expansion”)

$$b(t) = e^{-t/t_s}, \quad 0 \leq t \leq t_f \quad (8)$$

- ④ Phase Field equation with de Sitter expansion

The Fourier de Sitter Field Equation

$$\partial_\eta^2 \chi_{\mathbf{k}} - \frac{1}{\eta} \partial_\eta \chi_{\mathbf{k}} + c_0^2 k^2 \chi_{\mathbf{k}} = 0 \quad (9)$$

- ⑤ Initial conditions from the continuity of $\chi_{\mathbf{k}}$ and $n_{\mathbf{k}} = -\frac{\hbar}{U} \partial_t \chi_{\mathbf{k}}$ at $t = 0$.

Methods - Physical Theory - Particle Production

- 1 Can represent phase and density amplitudes in same basis

$$u_k(t) = \frac{1}{2\sqrt{n_0}} n_k(t) + i\sqrt{n_0} \chi_k(t) \quad (10a)$$

$$v_k(t) = \frac{1}{2\sqrt{n_0}} n_k(t) - i\sqrt{n_0} \chi_k(t), \quad (10b)$$

- 2 Find $u_k(t)$ and $v_k(t)$ out of and during the expansion

Particle Production

$$N_k(t) = \left| u_k^{\text{out}*}(t) v_k^{\text{exp}*}(t) - v_k^{\text{out}*}(t) u_k^{\text{exp}*}(t) \right|^2 \quad (11)$$

Methods - Physical Theory - Particle Production Algorithm

- 1 Solve the Fourier de Sitter field equation (9) for $\chi_{\mathbf{k}}(t)$ at a point in space $\mathbf{x}$
- 2 Differentiate $\chi_{\mathbf{k}}(t)$ to get $n_{\mathbf{k}}(t)$
- 3 Combine $\chi_{\mathbf{k}}(t)$ and $n_{\mathbf{k}}(t)$ to get the mixed Fourier amplitudes as in equations (10)
- 4 Calculate the number of particles at position $\mathbf{x}$ at time t with wave vector $\mathbf{k}$ as in equation (11)

Methods - Physical Theory - Computational Parameters

	Previous	New
Universe Size $\mathbf{x}$	128×128	$128 \times 128 \times 128$
Time Scale t_s	10^{-5} to 10^{-3}	10^{-5} to 10^{-3}
Wave Vector Domain $\mathbf{k}$	$\ \mathbf{k}\ \leq 32 \times 2\pi$	$\ \mathbf{k}\ < k_c$

Table: Previous and new computational domains

- This system only valid for $\|\mathbf{k}\| < k_c$ such that $\hbar^2 k_c^2 / 2m = Un_0$
- Time step such that the total particle number for the entire field changed by much less than one, or that the change in norm was small enough i.e. $\Delta t : \Delta \|\psi(\mathbf{x}, t)\| \leq 10^{-9}$

Methods - Computational Methods - Algorithms & Hardware

Physics	Method
Fourier Field Equation	4-th order Runge-Kutta
Density-Phase Transformation	4-th order explicit central FD
Particle Production	vectorized arithmetic

Table: Computational methods

Use the “Turing” supercomputer for its

- “tight”-cluster configuration
- dedicated head, storage, and several compute nodes
- NVIDIA GPUs allowing for CUDA massive parallel computing

Methods - Computational Methods - Programming Language

- Will choose between Mathematica or Julia programming languages considering the following:

	Experience	Open-source	Free	Scientific	Parallel	Cluster	GPU
Mathematica	X			X	X	X	X
Julia		X	X	X	X	?	X

Table: Language comparison

- Other high-performance languages like C++ and Java could also be considered if performance is needed
- Mathematica is not good for reproducibility

Timeline Overview

- ① Performance Scaling Phase - Summer '23
- ② Domain Scaling Phase - Fall '23
- ③ Writing Phase - Winter '24
- ④ Dissemination Phase - Spring '24

Timeline - Performance Scaling Phase - Summer '23

- 1 Convert hybrid analytic-numerical algorithms to fully numeric low-performance algorithms (LPNA) in final language of choice
- 2 Test the LPNAs in the 2D model and compare against previous results
- 3 Translating the LPNAs to high-performance fully-numerical algorithms (HPNA) that make use of high-performance hardware
- 4 Test the HPNAs in the 2D model and compare against previous results, measuring the efficiency when compared to the LPNA

Timeline - Domain Scaling Phase - Fall '23

- ① Convert 2D theory to 3D
- ② If not generalized, convert the 2D HPNAs to 3D
- ③ Test the 3D HPNAs on a small spatial domain and small momentum domain
- ④ Test the 3D HPNAs on a small spatial domain and large momentum domain
- ⑤ Do the “real” run of the 3D HPNAs on a large spatial domain and large momentum domain
- ⑥ Revise and repeat runs until satisfactory results are achieved.

Timeline - Writing & Dissemination Phases - Winter & Spring '24

① Writing Phase - Winter '24

- ① Write report
- ② Prepare for conferences

② Dissemination Phase - Spring '24

- ① Defend thesis
- ② Present at conferences
- ③ Submit final document to be considered for journal publication and graduation

Anticipated Results

Refer back to goals

- Insight into particle production during the moments just after the Big Bang
- Quantitatively describe the error in lower-dimensional models
- Quantitatively describe the worth of using full-dimensional model with high-performance computing

Anticipated Results - Particle Production Insight

- 2D and 3D field equations are qualitatively equal

$$\text{2D: } \partial_t^2 \theta - \frac{3}{2} \frac{b'(t)}{b(t)} \partial_t \theta - c_0^2 b(t) \nabla^2 \theta = 0 \quad (12a)$$

$$\text{3D: } \partial_t^2 \theta - \frac{b'(t)}{b(t)} \partial_t \theta - c_0^2 b(t) \nabla^2 \theta = 0 \quad (12b)$$

- I expect qualitatively similar results from previous research

Anticipated Results - Low-dimensional Model Error

- Qualitatively similar results should yield small error
- Will determine how small
- Small enough to not warrant future research using HPC based 3D model?
- Big enough that more accurate results are sometimes warranted?

Anticipated Results - Efficiency

- Recreated analytic 2D results

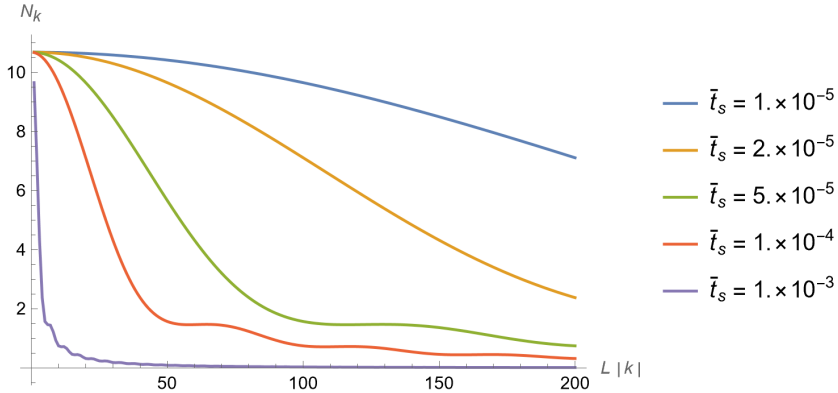


Figure: The momentum distribution when the universe stops expanding for multiple effective rates of expansion

Anticipated Results - Efficiency

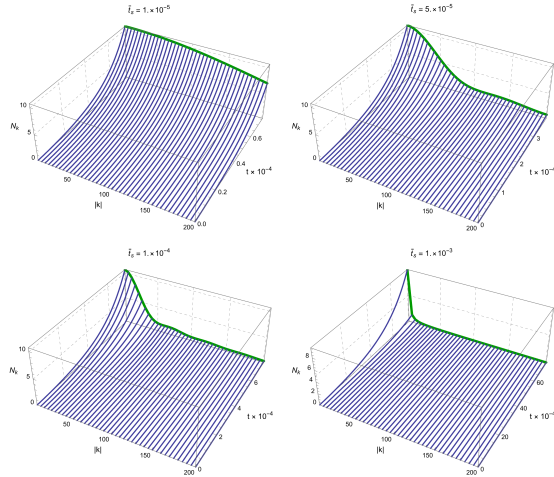


Figure: Momentum distribution throughout the expansion

Anticipated Results - Efficiency

- Used analytic solution for particle production
- Calculations still took a few seconds
- I expect the worth to be dependant on the domains of the parameters e.g. large spatial grid over a long time needs HPC but otherwise modern technology will yield resonable calculation times

Dissemination

- Journal-ready thesis submitted to Physical Review D
- Present at SOURCE
- Present at APS Northwest
- Possibly present at APS April Meeting

Thank you & Code source

Thank you for your attention.

The code for this project is available at:

- GitHub (QR code): github.com/nondairyneutrino/thesis
- My website: nathanwchapman.com



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